

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Department of Geology and Geophysics

2. Subject Name: Economic Geology

L-T-P: 3-0-0

Credits:3

3. Pre-requisites: Introduction to Earth Sciences

4. Syllabus and reference books:

Syllabus:

Spatial diversity of occurrence of Mineral Deposits in the crust in rocks ranging the entire Geologic history and in diverse tectonic domains, Important metallogenic provinces of the world; Ore Mineralogy, Morphologies and geometries of ore bodies in undeformed and deformed terranes; Various endogenic and exogenic processes of Mineral Deposit Formation; Important genetic types of ferrous, non-ferrous, precious, rare metals and rare earth element deposits; Role of Mineral resources in national economy, history of exploitation of mineral resources, elements of Mineral Economics; Classification schemes of world mineral resources, National Mineral Policies and Law of the Sea; Environmental Impacts of mineral resource exploitation.

Reference Books:

- 1) Economic Mineral Deposits by Jensen and Bateman (Wiley)
- 2) Introduction to Ore Forming Processes by Laurence Robb (Wiley)
- 3) Ore Geology and Industrial Minerals by Anthony Evans (Blackwell)

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Introductory Concepts of Metallogeny	3
2.	Ore Mineralogy	3
3.	Mineral deposit morphologies and their controls	3
4.	Endogenic and Exogenic processes	15
5	Mineral Economics	9
6	Environmental Impacts of Mineral Resource Exploitation	3
Total number of hours		36